

Instructor Notes:



Instructor Notes:**Lesson Objectives**

In this lesson, you will learn about:

- Array object
- Properties and methods of Array objects



Instructor Notes:

4.1: Array Object

Concept of Array Objects

An array is the sole JavaScript data structure provided for storing and manipulating ordered collections of data

For creating an array, you can use the following

```
var myArray = new Array() //empty array
```

```
solarSys = new Array(2) //Array defined with size  
solarSys[0] = "Mercury" // Assigning values to array  
solarSys[1] = "Venus"
```

```
solarSys = new Array("Mercury", "Venus", ...) condensed  
array
```

```
solarsys=["Mercury","Venus",....] // literal array
```

Array Objects:

An array is a kind of variable that can hold more than one value at a time.

However, unlike some other programming languages, JavaScript's arrays are very forgiving as to the kind of data you store in each cell or entry of the array. This allows, for example, an array of arrays, providing the equivalent of multidimensional arrays customized to the kind of data your application needs.

You can see a few examples listed on the slide for creating an array. We see example of creating an empty array. To limit the size of array you can specify the optional integer value as seen in the second example.

Another way of defining an array is called as condensed array which allows you to combine the array and array elements definitions into one step.

Literal arrays are define by assigning the value in square brackets. To create an array with initial undefined values you can simple enter a comma. For eg

```
myarray=["Pune",,, "Mumbai"]
```

Instructor Notes:

4.2: Properties and methods of Array Object

Concept of Array Object Methods

JavaScript provides the following array object methods:

- `arrayObject.length`
- `arrayObject.join(separatorString)`
- `arrayObject.reverse()`

Example for length method

- `myArray.length` // result: 5

Code snippet for usage of join method

- In this, `myArray` contents will be joined and placed into `arrayText` by using the comma separator

```
var arrayText = myArray.join(",")
```

Array Object Methods:

After you have information stored in an array, JavaScript provides several methods to help you manage that data.

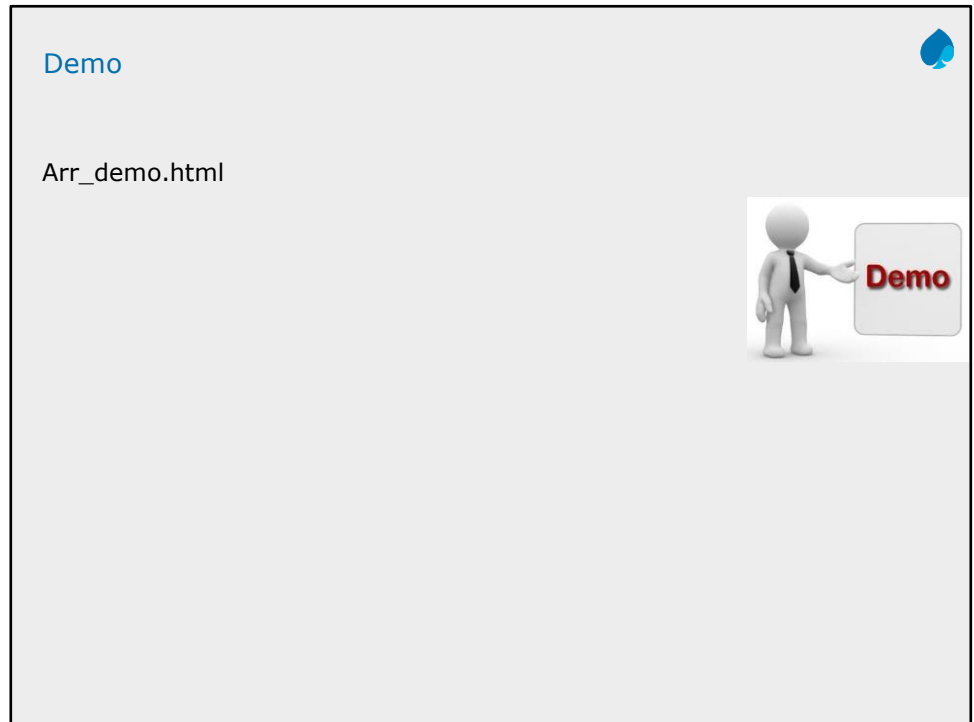
`arrayObject.join(separatorString)`

It returns string of entries from the array delimited by the `separatorString` value.

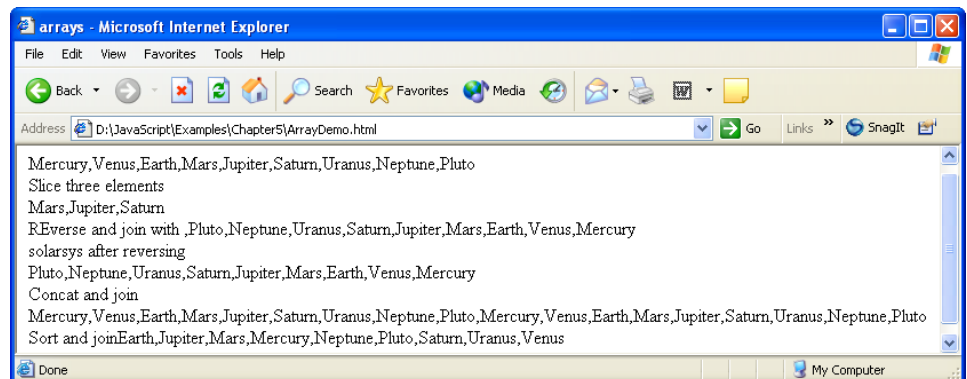
```
var arrayText = myArray.join(",")
```

It returns array of entries in the op.

The element that was last in the array becomes the 0 index item in the array. Note that when you do this, you are restructuring the original array, and not copying it.

Instructor Notes:

Demo of Array object (Arr_Demo.html) produces the output as shown below:



Instructor Notes:**Lab****Lab 4 :**

- Working with Arrays



Instructor Notes:**Summary**

An array is a set of variables
You can create an array object by using new operator
Array Object properties are length
Array Object Methods are concat, join, reverse, slice etc

**Summary**

Instructor Notes:**Review Question**

Question 1: The ____ method allows you to join array contents and place it into a text

- Option 1: array.concat()
- Option 2: array.join()

Question 2: An array object automatically has a size property.

- True/False

